Presentation and Handout Guidelines: Data Analysis for Decision-Making (WS 2026)

Prof. Dr. Stephan Huber (Matr. Nr. 1234567890)

Fresenius University of Applied Science

Data Analysis for Decision-Making

Prof. Dr. Stephan Huber

Author Note

Correspondence concerning this article should be addressed to Prof. Dr. Stephan Huber (Matr. Nr. 1234567890), Fresenius University of Applied Science, Email: stephan.huber@hs-fresenius.de

Abstract

This document outlines the requirements for successfully contributing to the course ‘Data Analysis for Decision-Making’. It introduces and assigns the available topics, establishes the standards for both delivery and the creation of files, and provides tips how students can efficiently and effectively meet the expectations.

Presentation and Handout Guidelines: Data Analysis for Decision-Making (WS 2026)

Assessment methods and criteria

Students complete this module with an academic presentation and a handout. The presentation will take place during the lecture period, with the exact dates set by the lecturer. Each presentation lasts a minimum of 10 minutes per student. Additionally, students should produce a handout outlining the key points of your presentation and the literature on which it is based. The handout should provide additional information and resources beyond what is covered in the presentation. In other words, the handout should complement the presentation. The length of the handout should be between 3 and 5 pages of text.

The presentation contributes 65% to the module grade, while the handout contributes 35%. A passing grade for this module is achieved when the overall grade is greater than or equal to 4.0.

Content of the presentation and handout

Your presentation should aim to enhance your fellow students' understanding of the assigned topic. Given the limited presentation time, it is essential to select your content carefully. One of your key challenges is to identify important elements of your topic and present them in a manner that can be communicated and understood within the allocated time. In some cases, your presentation may also ignite the audience's curiosity, encouraging them to explore the topic further on their own.

While you are responsible for the overall content, your submission must include R code and an exercise related to your topic. The exercise should be completed by the students during the class immediately following your presentation.

Form of the presentation and the handout

Create a presentation and a handout using Quarto. The handout layout should adhere to the style guide of APA (2020). Please use the Quarto extension `apaquarto`, which you can find here: https://github.com/wjschne/apaquarto. Alternatively, you can clone the repo https://github.com/hubchev/dadm_ws26 and use the QMD file that you find there as a template.[1]

[1] Clone the repo by typing the following in your terminal: `git clone https://github.com/hubchev/dadm_ws26`

Both your handout and presentation should be available online on the day of your presentation. Ensure the HTML file is a standalone website (HTML), and also provide a PDF version of your handout and presentation. The PDF of your handout should be formatted for A4 paper. For guidance on creating a standalone HTML file, refer to Quarto HTML Publishing. Publish both the website and handout on GitHub, sharing the URLs with the audience for access during your presentation. Additionally, please submit at least the following files on ILIAS:

- Quarto files of your handout and presentation.
- HTML files of your handout and presentation.
- PDF files of your handout and presentation.

The design and the layout of the presentation slides are your choice. However, please avoid trying to impress with elaborate layouts or extraneous details. This is an academic presentation, and distracting decorations are inappropriate. Your primary focus should be on effectively communicating information, facts, and insights to the reader. Feel free to include any elements that support this goal. In the presentation, tables and figures don't need to be numbered. Also, consider additional requirements for your submission outlined below.

Below you see an affidavit. Fill it out and put it at the end of your report.

Affidavit

I hereby affirm that this submitted paper was authored unaided and solely by me. Additionally, no other sources than those in the reference list were used. Parts of this paper, including tables and figures, that have been taken either verbatim or analogously from other works have in each case been properly cited with regard to their origin and authorship. This paper either in parts or in its entirety, be it in the same or similar form, has not been submitted to any other examination board and has not been published.

I acknowledge that the university may use plagiarism detection software to check my thesis. I agree to cooperate with any investigation of suspected plagiarism and to provide any additional information or evidence requested by the university.

Checklist:

- ☐ My handout contains 3-5 pages of text (The front page, the affidavit, and the reference list do not count).
- ☐ My submission contains the Quarto file of the handout.
- ☐ My submission contains the Quarto file of the presentation.
- ☐ My submission contains the HTML file of the handout.
- ☐ My submission contains the HTML file of the presentation.
- ☐ My submission contains the PDF file of the handout.
- ☐ My submission contains the PDF file of the presentation.
- ☐ The title page of my presentation and handout contain personal details (name, email, matriculation number).
- ☐ My handout contains a bibliography, created using BibTeX with an APA citation style.
- ☐ Either the handout or the presentation contains R code that verifies my expertise in coding.
- ☐ The Affidavit is filled out correctly.
- ☐ The PDF and HTML files of the presentation and the handout are published on GitHub Pages.
- ☐ I submitted a pull request to the repository https://github.com/hubchev/make_a_pull_request, where I added my GitHub username and the repository related to this submission to the file https://github.com/hubchev/make_a_pull_request/dadm_ws26.md.

[Your Name,] [Date,] [Place]

Tip

- You can check the box in Quarto like this:

```
-   [x] I checked this box
```

- A template for the handout can be found here: https://github.com/hubchev/temp_apa_stu

General tips

I refrain from specific advice on good writing, structuring your work, or adhering to academic rules and conventions. Therefore, I recommend reading Huber (2025) and the literature cited therein.

Since this is an academic presentation and work, respectively, please choose your sources carefully and ensure that you cite them appropriately. If anything is unclear, please let me know as soon as possible. It is your responsibility to be well prepared on the day of your presentation. If you encounter any problems, let me know immediately and I will do my best to help you. However, please understand that I may have limited availability in the days or weeks leading up to your presentation. Prioritize the essential tasks that are technically complex or where something is likely to go wrong.

Topics and dates

Please consider Table 1. You should be able to identify yourself with your first name.

! Important

- If you do not see your name on the list but have registered, please contact me as soon as possible.
- The presentation day is an official examination day, and failure to attend will result in a grade of 5.0. Ensure that you do not miss this important date.

Table 1

Topics, students, and dates

Date	Name	Topic
Nov 18	Moura	Literate Programming With Quarto[2]
Nov 18	Bani	Importing Data in R: Exploring Various Packages and Their Advantages[3]
Nov 18	Yang	Version Control and RStudio[4]
Nov 25	Azimi	Writing R Code with NextGenAI Assistance
Nov 25	Wang	R Control Flow Statements[5]
Nov 25	Schenk	R Control Flow Statements and Flow Charts[6]
Dec 2	Dai	Time Series Visualization in R
Dec 2	Virda	The Importance of Summary Statistics
Dec 2	Gokhale	Accessing Internet Data Sources in R with Convenient Packages
Dec 2	Khatkar	Descriptive Tables in R with dplyr and gtsummary
Dec 9	Cayllahua	Why Data Cleaning Matters: Methods and Tools in R[7]
Dec 9	Stefiany	Handling Duplicates in Data: Challenges and Solutions[8]
Dec 9	Rabiee	Interactive Visualization in R & Applications in Business
Dec 9	Kanojia	Interactive Business Analytics Using R and Shiny: The Radiant Package
Dec 16	Balaguru	Geospatial Visualization with R
Dec 16	Wu	Applying Cluster Analysis in R for Business Segmentation
Dec 16	Gurbanov	Factor Analysis for Business Decision-Making Using R
Dec 16	Wayuphat	Visualizing Multiple Variables With R
Dec 16	Boda	Conducting Multiple Linear Regression Analysis Using R

[2]Quarto is a good application to build reproducible research and produce publications. Explain *literare programming* and focus on how to include code in Quarto. See: Rodrigues (2023), carpentries-incubator (2025)

[3]See: Wickham et al. (2025, ch. 7), Batra et al. (2024, ch. 7)

[4]See: Batra et al. (2024, ch. 46)

[5]Please collaborate with Dai.

[6]Please collaborate with Cayllahua.

[7]Please collaborate with Rabiee

[8]Please collaborate with Moura

Date	Name	Topic
Dec 16	Tchakoutio	Understanding Syntax: Python and R Compared

References

APA. (2020). *Publication manual of the American Psychological Association* (7th ed.). American Psychological Association.

Batra, N., Spina, A., Blomquist, P., Campbell, F., Laurenson-Schafer, H., Florence, I., Fischer, N., Ndiaye, A., Coyer, L., Polonsky, J., Izawa, Y., Bailey, C., Molling, D., Berry, I., Buajitti, E., Mousset, M., Hollis, S., & Lin, W. (2024). *The epidemiologist R handbook* (N. Batra, Ed.). https://www.epirhandbook.com

carpentries-incubator. (2025). *Introduction to reproducible publications with Quarto*. https://carpentries-incubator.github.io/reproducible-publications-quarto/aio.html

Huber, S. (2025). *My five cents on how to write a thesis: A guide for my students*. https://hubchev.github.io/fivecents/fivecents.pdf

Rodrigues, B. (2023). *Building reproducible analytical pipelines with R*. https://raps-with-r.dev

Wickham, H., Çetinkaya-Runde, M., & Grolemund, G. (2025). *R for data science (2e): Import, tidy, transform, visualize, and model data*. online. https://r4ds.hadley.nz/